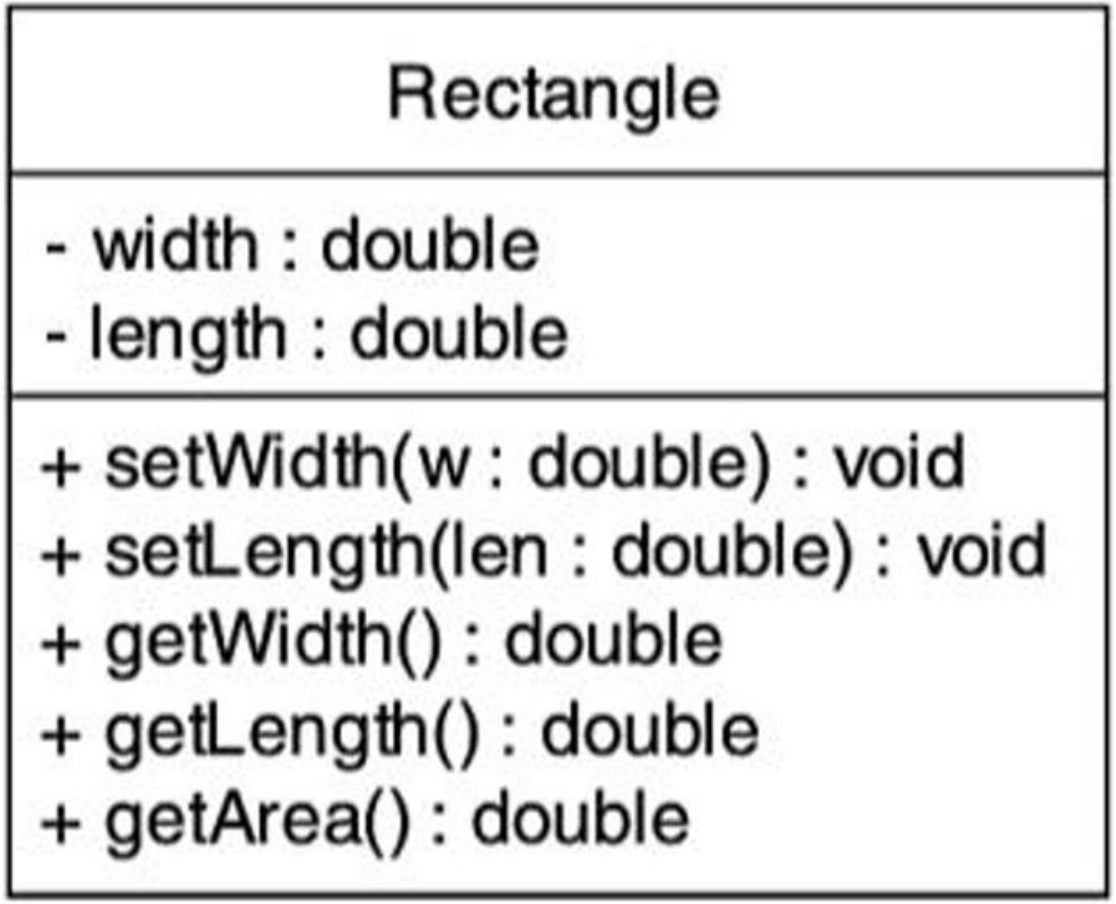
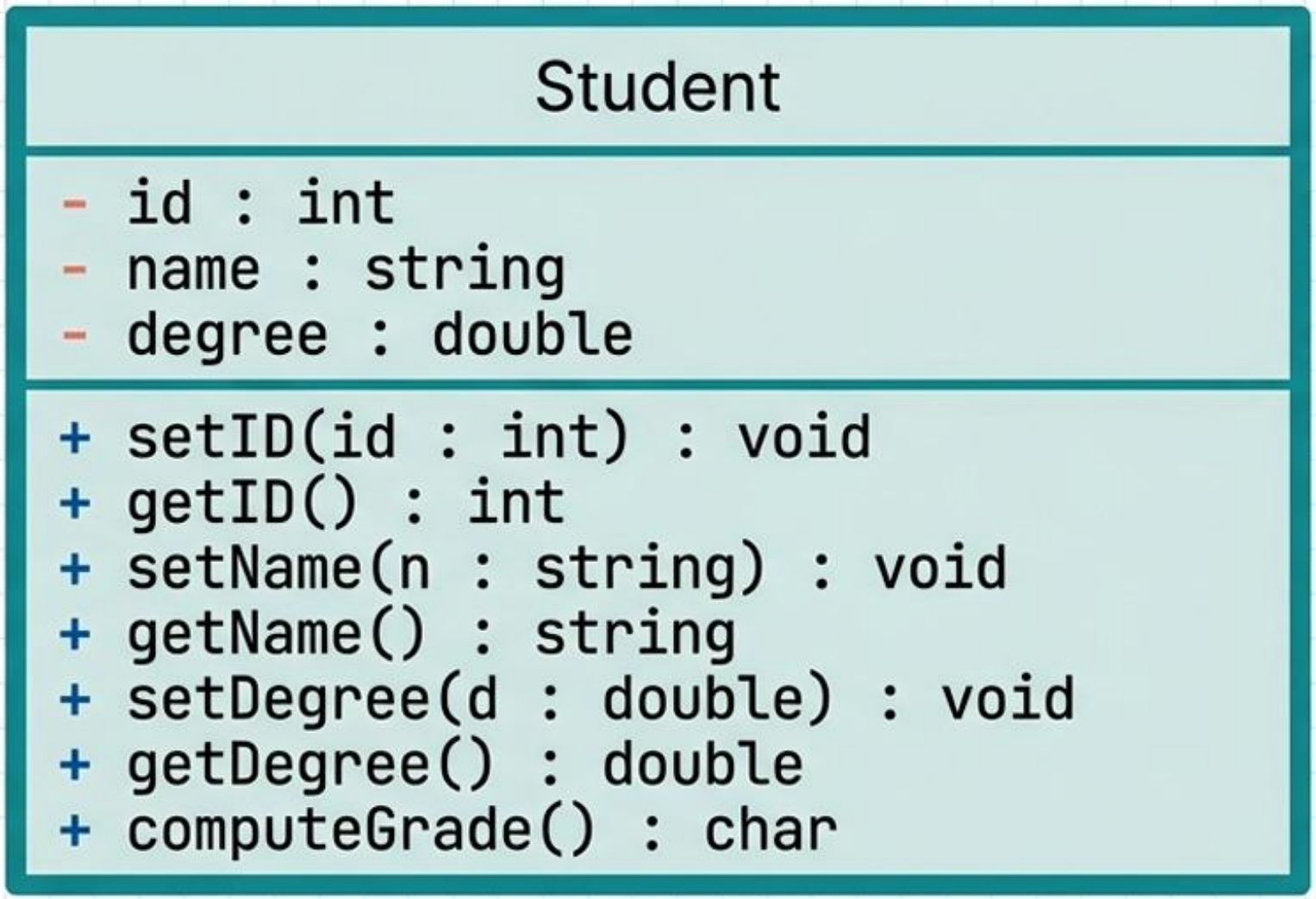
Lab1 and 2

- Convert the following UML to the corresponding class.
- Be sure to apply abstraction aspect.
- Divide the class to .h and .cpp
- Test the class into main.cpp




Student
- id : int
- name : string
- degree : double
+ setID(id : int) : void
+ getID() : int
+ setName(n : string) : void
+ getName() : string
+ setDegree(d : double) : void
+ getDegree() : double
+ computeGrade() : char

Applied Practice: E-Commerce Product

```
Product
- price : double
- stockCount : int

+ buy(quantity : int) : void
+ restock(quantity : int) : void
+ getPrice() : double
```

- **Public** methods enforce logic (checking stock) before modifying private data.

Applied Practice: Bank Account

Sensitive financial data is locked. Note the private helper method logTransaction.

BankAccount

```
- accountNumber : string
- balance : double
```

```
+ deposit(amount : double) : void
+ withdraw(amount : double) : boolean
- logTransaction() : void
```

Applied Practice: Library Book

Book

- isbn : string
- isBorrowed : boolean

+ borrow() : boolean
+ returnBook() : void
+ isAvailable() : boolean